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DIFFERENTIATION AND LOCAL TUMOR RESPONSE IN
RADIATION THERAPY OF LIMITED NON-OAT CELL
CARCINOMA OF THE LUNG

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Among 21 patients with limited non-oat cell carcinoma of the lung, 9 patients had well or moderately well differentiated (grade 1 or 2) epidermoid carcinoma or adenocarcinoma, and 12 had poorly differentiated (grade 3 or 4) epidermoid carcinoma or adenocarcinoma, or large cell undifferentiated carcinoma. They were treated by definitive radiation therapy with 4500 rads in 180 to 200-rad fractions five times per week to primary tumor, mediastinum and supraclavicular nodes. Following this large field treatment, small field treatment was given to radiographically apparent residual tumor to complete response or 7000 rads (for residual lesions under 4 cm largest diameter) or 6500 rads (for larger lesions). Response of primary tumor was determined by chest x-ray at approximately 4500 rads, 5500 rads, and 6500 rads, as well as at a follow-up examination 9-12 weeks after start of radiotherapy. Cumulative local tumor response was determined, and CR indicates complete radiographic disappearance of lesions; PR indicates more than 50% reduction in surface area on PA chest x-ray:

Dose (x 1000 rads)		4.0-5.0	4.0-6.0	4.0-6.5	4.0-7.0	Follow-up
Well differentiated	CR	0%	11%	11%	22%	22%
	PR	33%	33%	33%	33%	44%
Poorly differentiated	CR	16%	33%	33%	40%	66%
	PR	24%	40%	40%	40%	25%

We conclude that poorly differentiated carcinomas show more rapid and more frequent local response to radiation therapy than well differentiated lesions. The effect of this increased local tumor response on survival remains to be determined.

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ADENOCARCINOMA OF THE LUNG
TEN YEARS' EXPERIENCE IN SOUTHERN ALBERTA CANCER CENTER

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During the period from 1968-1977, 1223 patients with carcinoma of the lung were registered at the Southern Alberta Cancer Center, of which 121 had adenocarcinoma of the lung. A detailed retrospective analysis was undertaken to